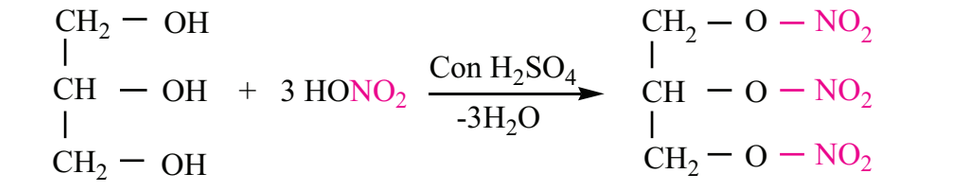
## Reaction of Glycerol

**Nitration:** Glycerol reacts with concentrated nitric acid in the presence of concentrated sulphuric acid to form TNG (nitroglycerine).

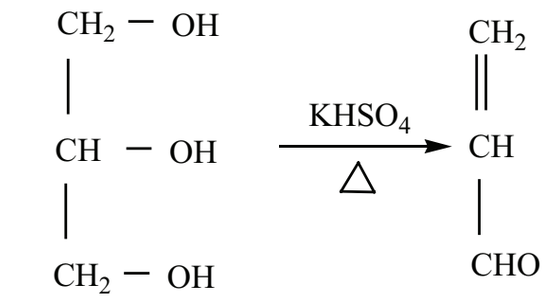



Propan - 1,2,3 - triol
glycerol

1,2,3 - trinitroxy propane

## Dehydration

When glycerol is heated with dehydrating agents such as Con $H_2SO_4$,$KHSO_4$ etc…., it undergoes dehydration to form acrolein.




Propane - 1,2,3 - triol

Prop - 2- enal (acrolein)

## Oxidation

Glycerol can give rise to a variety of oxidation products depending on the nature of the oxidising agent used for oxidation.

a) Oxidation of glycerol with dil. $HNO_3$ gives glyceric acid and tartronic acid.

b) Oxidation of glycerol with Conc. $HNO_3$ gives mainly glyceric acid.

c) Oxidation of glycerol with bismuth nitrate gives as meso oxalic acid.

d) Oxidation of glycerol with $Br_2/H_2O$ (or) NaOBr (or) Fenton's reagent ($FeSO_4 + H_2O_2$) gives a mixture of glyceraldehyde and dihydroxy acetone(This mixture is named as glycerose).

e) On oxidation with $HIO_4$ or Lead tetra acetate (LTA) it gives formaldehyde and formic acid.

f) Acidified $KMnO_4$ oxidises glycerol into oxalicacid.



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